

- The **colon** extends from the caecum up the right side of the abdomen, across the upper abdomen, and then down the left side of the abdomen, finally connecting to the rectum. The colon has three parts: the ascending colon and transverse colon, which absorb fluids and salts, and the descending colon, which holds the resulting waste (faeces). Faeces mainly consist of undigested material, large number of bacteria, sloughed off gastrointestinal cells, bile pigments and water. Bacteria in the colon help to digest the remaining food products.
- The rectum is where faeces are stored until they leave the digestive system through the anus as a bowel movement.

Liver and its functions:

The liver produces bile, which helps the body to digest and absorb fat. Bile is stored in the gallbladder until it is needed. Bile travels through special channel (bile duct) directly into the small intestine.

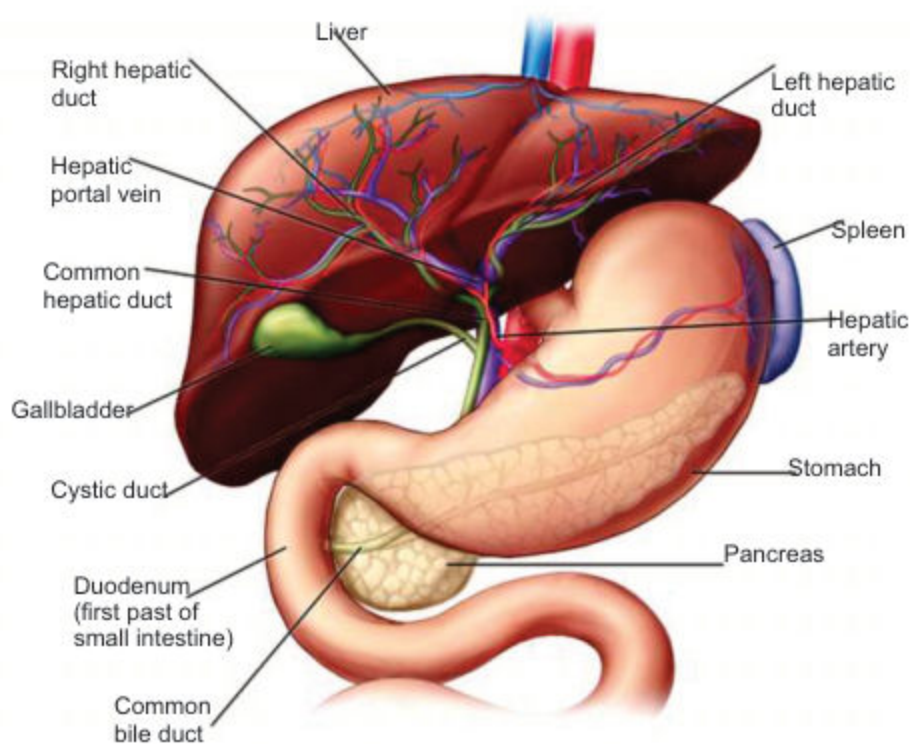


Figure 8.21 The human liver

It also makes a substance that neutralizes stomach acid. The liver also plays a major role in the handling and processing of nutrients, which are carried to the liver in the blood from the small intestine.

The liver is a metabolically active organ responsible for many vital life functions.

8.4 DISORDERS OF GUT

1. Diarrhea:

It is a condition in which the sufferer has frequent watery, loose bowel movements due to fast peristaltic movement. This condition may be accompanied by painful abdominal cramps, nausea, fever and generalized weakness. It occurs when required water is not absorbed in blood from colon. The main cause of diarrhea includes lack of adequate safe water, virus and bacteria. In malnourished individuals, diarrhoea leads to severe dehydration and can be life threatening. To control diarrhea consume adequate amounts of water to replace loss, preferably mixed with essential salts and some amount of nutrients.

2. Constipation:

Constipation is a condition, where a person experiences hard faeces that are difficult to eliminate. The main causes of constipation are hardening of faeces due to excessive absorption of water through colon, insufficient intake of dietary fibre, dehydration, use of medicine (e.g. those containing iron, calcium and aluminum) and tumors in rectum or anus. The treatment of constipation are change in diet and exercise habits, use of laxative (e.g. paraffin) may be in some cases. To prevent Constipation is easier than treatment.

3. Ulcer (peptic ulcer):

It is a sore in gut lining and can be different organs such as; ulcer of stomach is called "gastric ulcer", Ulcer of duodenum is called "duodenal ulcer", ulcer of oesophagus is called "esophageal ulcer" and breakdown of tissues by acidic gastric juice. It can be due to the long term use of anti-inflammatory medicine (e.g. aspirin), smoking, drinking coffee, colas and eating spicy food. Few of the signs are as under; abdominal burning after

meals, abdominal pain, rush of saliva after an episode of regurgitation, nausea and loss of appetite and weight. Ulcer can be treated with medicine, containing (alkaline composition) and avoiding spicy food.

Summary

- Process by which organism obtain and use the nutrients required for maintaining life is called nutrition.
- Autotrophic nutrition and heterotrophic nutrition
- Autotrophic nutrition found in plants and some bacteria specially photosynthesis.
- Heterotrophic nutrition found in animal and fungi which gets nutrients from other sources.
- Heterotrophic which use plant as food called herbivorous and those which consume animals. and their products called carnivorous. Both are collectively called consumers.
- The basis of mode living and the mode of intake of food heterotrophes may be parasitic, saprotrophic and holozoic.
- Nutrition in unicellular organisms, like amoeba involve ingestion by cell surface.
- Process involving the absorption, distribution and utilization of mineral substances by plant called mineral nutrition.
- The mineral nutrients are macronutrient i.e require in large quantity, micronutrient i.e require in small quantity.
- Fertilizers are substances containing chemical elements such as manure or mixture of nitrates that improve growth of plants.
- Naturally occurring materials which are not chemically modified called inorganic fertilizer.
- Chemical substances which are more complex and takes time to be broken down into useable form called organic fertilizer.

- So many environmental hazard are also related to chemical fertilizers
- There are seven major classes of nutrients, Carbohydrates, proteins, fats, minerals, fibers, vitamins and water.
- Balanced diet is related to age, sex and activity of human.
- Problems related to nutrition are grouped as malnutrition.
- Kwashiorkor is a severe difficulty in dietary protein.
- Marasmus characterized by energy deficiency sum of mineral difficulty diseases are Goiter, anemia.
- Effects of malnutrition are starvation, heart diseases, constipation, obesity.
- Digestion is the break down of complex food into simple absorbable nutrients.

Review Questions

1. Encircle the correct answer:

- Select the mismatched
 - Protein → Amino acid
 - Carbohydrate → Glucose
 - Fats → Starch
 - Nucleic acid → Nucleotide
- Deficiency of vitamin-k cause
 - Rickets
 - Anemia
 - Scurvy
 - Beriberi
- Period of rapid growth and development called
 - Childhood
 - Adult hood
 - Adolescence
 - Both "a" and "b"
- Deficiency of dietary protein, causes an imbalance in the gastro-intestinal system
 - Marasmus
 - Edema
 - Diarrhea
 - Kwashiorkor

- (v) An animal that migrates great distances would obtain the greatest benefit from storing its energy as
- (a) Fats
 - (b) Carbohydrates
 - (c) Protein
 - (d) Minerals.
- (vi) Which of the following vitamins is correctly associated with its use?
- (a) Vitamin K → Production of white blood cells
 - (b) Vitamin C → Curing rickets
 - (c) Vitamin E → Protection of skin from cancer
 - (d) Vitamin A → Incorporated into the visual pigment of the eye
- (vii) Which of the following statements describes pepsin?
- (a) It is manufactured by the pancreas.
 - (b) It helps stabilize fat-water emulsions.
 - (c) It splits maltose into monosaccharides.
 - (d) It begins the hydrolysis of proteins in the stomach.
- (viii) Which of the following is true of bile salts?
- (a) They are enzymes
 - (b) They are manufactured by the pancreas
 - (c) They emulsify fats in the duodenum
 - (d) They increase the efficiency of pepsin action.
- (ix) In human digestive system trachea and oesophagus both connect to the
- (a) Large intestine
 - (b) Stomach
 - (c) Pharynx
 - (d) Rectum
- (x) All are sources of calcium except
- (a) Red meat
 - (b) Green leafy vegetables
 - (c) Broccoli
 - (d) Nuts

2. Fill in the blanks:

- (i) Sea food is an excellent source of protein because it's usually low in_____.
- (ii) Iron deficiency is associated with decreased cognitive abilities and resistance to_____.
- (iii) Wave of rhythmic movements of muscle contraction and relaxation called_____.
- (iv) Liquid secreted by the pancreas, which contains a variety of _____.
- (v) Living organisms which derive their food from dead and decaying organic materials called_____.
- (vi) Nutrients that are needed in the highest concentration called_____.
- (vii) Fertilizers are substances containing chemical elements such as manure or mixture of _____.
- (viii) High solubility of fertilizers also degrades ecosystem through_____.
- (ix) Carbohydrates which digested quickly are called _____.
- (x) Each gram fat supplies the body with about_____.

3. Define the following terms

- | | | |
|------------------|-------------------|--------------|
| (i) Vitamins | (ii) Malnutrition | (iii) Goiter |
| (iv) Anemia | (v) Constipation | (vi) Obesity |
| (vii) Starvation | (viii) Ingestion | (ix) Chyme |
| (x) Ulcer | | |

4. Distinguish between the following in tabulated form.

- (i) Fat-soluble vitamins and water soluble vitamins
- (ii) Marasmus and kwashiorkor
- (iii) Chemical digestion and mechanical digestion
- (iv) Autotrophic nutrition and heterotrophic nutrition
- (v) Inorganic fertilizers and organic fertilizers

5. Write short answers of following questions.

- (i) Why fertilizers are necessary for plant?
- (ii) How stomach linings are protected from acidic environment?
- (iii) Why nitrogen is essential for plants?
- (iv) Why fats are called most efficient form of food?
- (v) Why grinding and lubrication are necessary for swallowing?

6. Write detailed answers of the following questions.

- (i) Describe environmental hazards related to chemical fertilizers.
- (ii) Describe function of stomach and intestine with suitable diagram of human digestive system.
- (iii) What are vitamins? Describe types of vitamins
- (iv) Describe mineral deficiency diseases in human.
- (v) Describe effects of malnutrition on human.

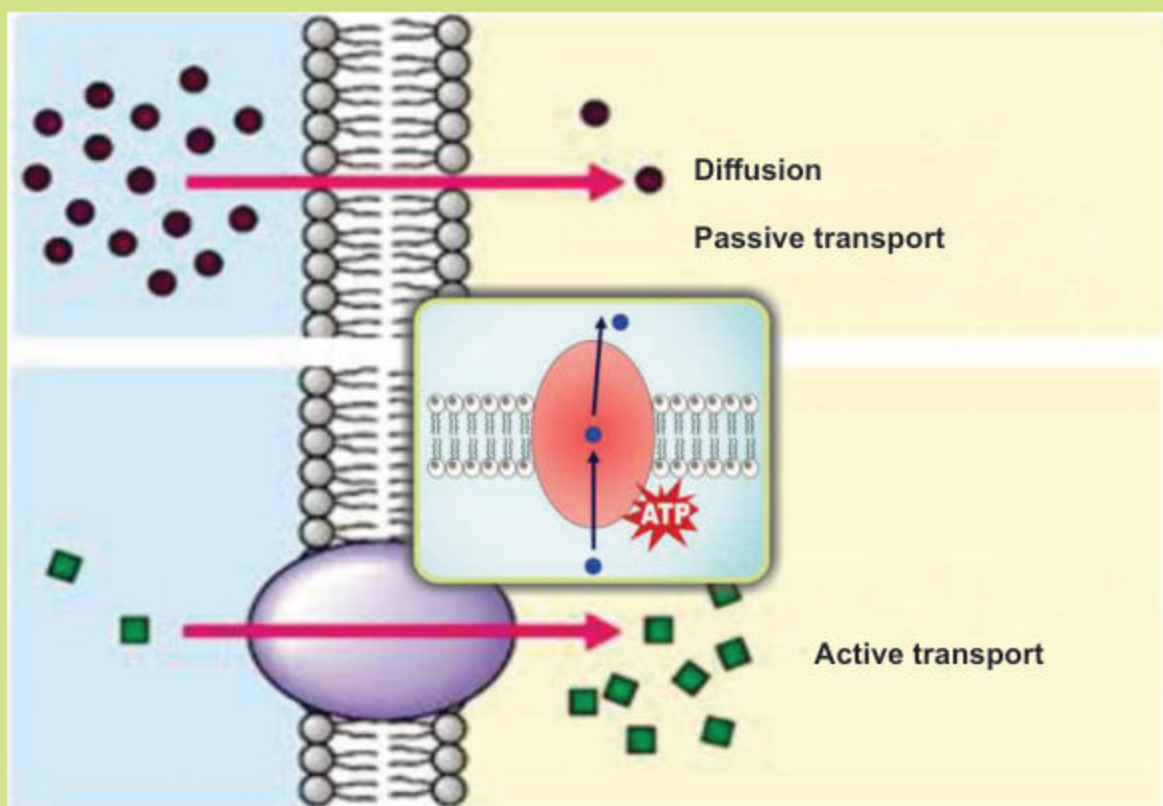
TRANSPORT

Chapter 9

Major Concept

In this Unit you will learn:

- Introduction
- Transport in Plants
 - Water and ion uptake (Structure and function of root hairs)
- Transpiration
 - Introduction and Significance
 - Factors affecting the rate of transpiration
- Transportation of Food and Water
 - Pathway of water and food in stem
- Structure and function of Xylem and Phloem
- Transport in Animals
- Transport in Man
 - Blood
 - Components of Blood and their Functions
 - Disorders of Blood (Leukemia and Thalassemia)
 - Blood Groups and Blood Transfusion
- Human heart
 - Blood Vessels



Every organism requires number of substances for their survival and maintenance of healthy life. These substances or their raw materials are taken by organism from environment or may be from internal sources. If the distance between source and required organ is small enough, organism does not require any transport system but if the distance is too long then the organism require to develop a system called **transport system**. The transport system requires atleast two component.

- (i) Transport of raw material from environment to organ where they required for metabolism.
- (ii) Transport of metabolites from cell to organs where they require.

Plants are autotrophs which synthesize organic biomolecules from inorganic molecules. These inorganic molecules are transported from environment into plants, converted into biomolecules. These biomolecules are also transported within the plant where they required.

Animals are heterotrophs which get organic molecules as food, digest them into digestive tract, diffuse into blood and transport to organs where required.

9.1 TRANSPORT IN PLANTS

Root as important organ for water and Mineral transport:

Water and mineral salts enter the plant through root, so it is necessary to understand the external and internal structure of root. Externally, root has a root tip which is the growing part of root covered by root cap. The remaining part of root is highly branched and each branch is heavily covered by root hairs. Each root hair is a fine tubular outgrowth of an epidermal cell. It grows between soil particles which remains in close contact with the soil solution surrounding them.

Internally, we can study the root by taking transverse section (T.S.) of root. The T.S. of root shows that root is mainly consist of:

- **Epidermis** (Epiblema) the outer layer of cells, some of these cells have root hair.
- The **Cortex** part of root between epidermis and endodermis, consist of number of cellular layers.

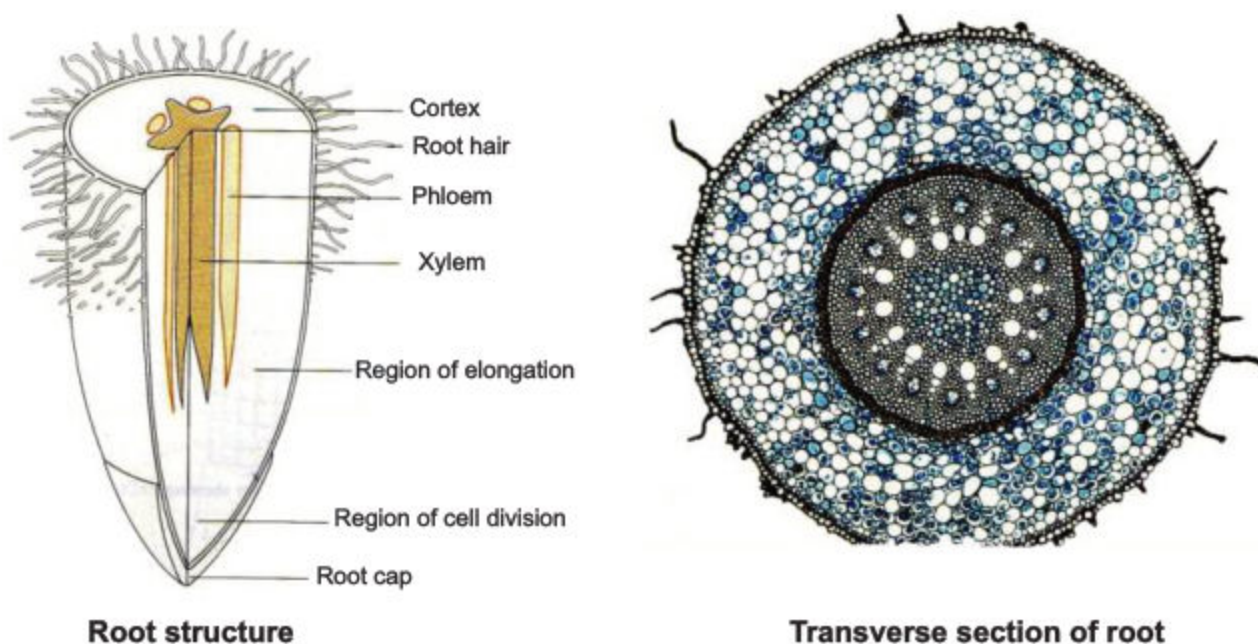


Fig: 9.1 Root structure with transverse section of root

9.1.1 Water and ions uptake:

The root absorb water and minerals from soil through root hairs. There are two processes of transport:

- Passive transport:** The uptake of water and mineral by osmosis and diffusion without using energy of ATP. It is due to concentration gradient i.e. always takes place from high to low quantity of substances.
- Active transport:** Movement of substances from low quantity to high quantity i.e. against the gradient and it requires energy of ATP. This movement is called active transport.

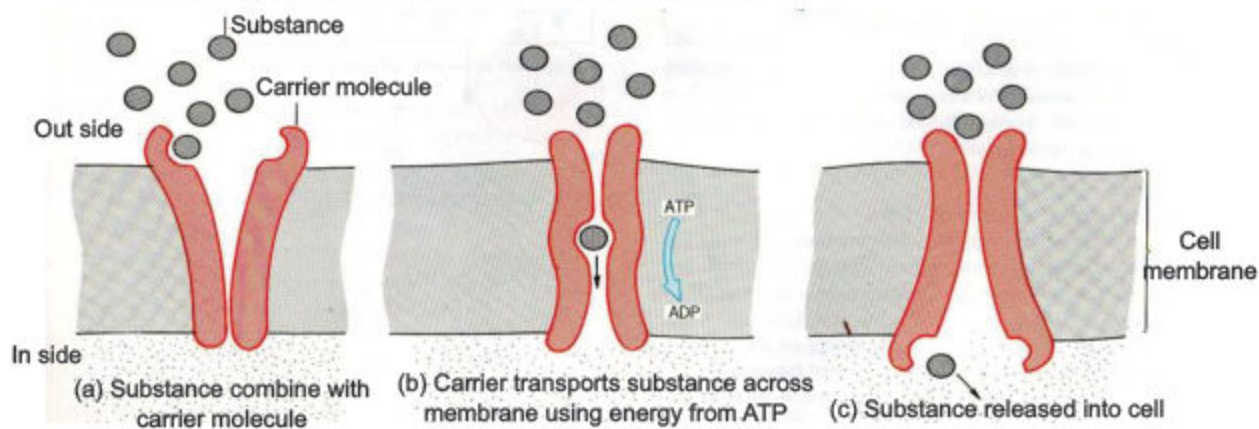


Fig: 9.2 Active transport

1. Uptake of Water From Soil:

The root hair is long, thin and tubular structure. It increases the surface area which increases the rate of absorption of water and minerals. The cell membrane prevents the cell sap (a mixture of sugar, salts and amino acids in solution form) from leaking out. The cell sap has low tendency of water loss i.e water potential, than the solution of soil which result in water movement from soil to root hair. This process of water movement from a solution of high water potential to a solution of low water potential is called **Osmosis**. As a result of osmosis of water, the root hair become turgid and their cell-sap become dilute than that of adjacent cells so the water moves from root hair to their neighbour cells. In this way, water may pass from cell to cell and finally forced into xylem and ascend up to the aerial part of plant. This upward movement of water with mineral is called ascent of sap, other factor and forces are also involved in ascent of sap.

For absorption of water by root, it is necessary that the solute quantity in soil solution should be low and solute quantity in sap should be comparatively high, otherwise the direction of water movement will be reversed and plant may die due to dehydration.

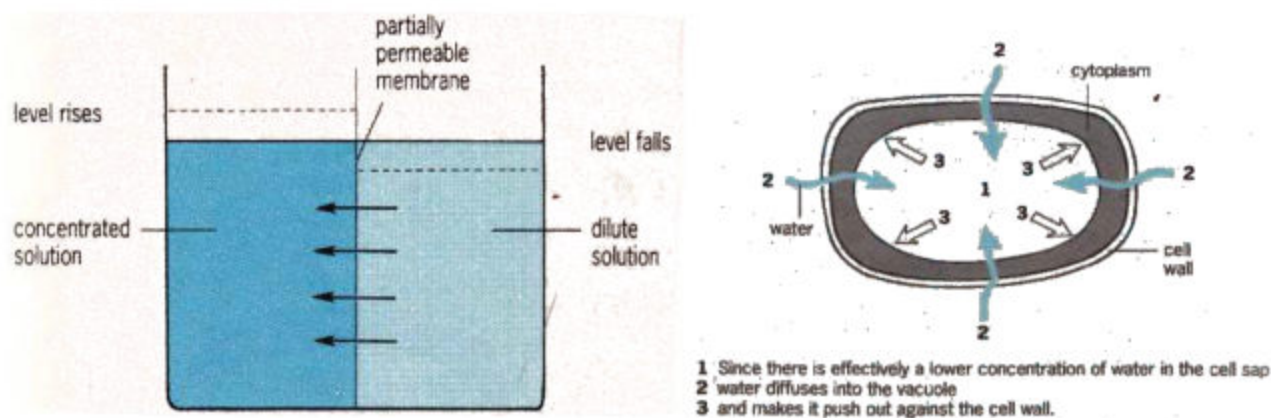


Fig: 9.3 Diagram showing process of Osmosis and Diffusion

2. Mineral transport:

Plant also require minerals i.e nitrates, sulphates, phosphates etc. These minerals are also taken up by root hair in two ways:

- (a) By diffusion, when the concentration of certain ions in soil is higher than that in root hair cells i.e passive transport.

- (b) By active transport, plant requires some substance even they found in soil in low quantity. The roots have to absorb these ions against a concentration gradient by using energy of ATP, which is active transport.

9.2 TRANSPIRATION

Plants absorb water continuously from soil. Some of its quantity utilized in photosynthesis and other metabolic functions while the rest is retained in cell to maintain turgidity of cell. Some water is removed in the form of vapours. This loss of internal water of plant in the form of vapours from aerial part of plant is called **transpiration**. Transpiration mainly takes place through special pores guarded by specialized guard cells called **stomata** (sing: stoma).

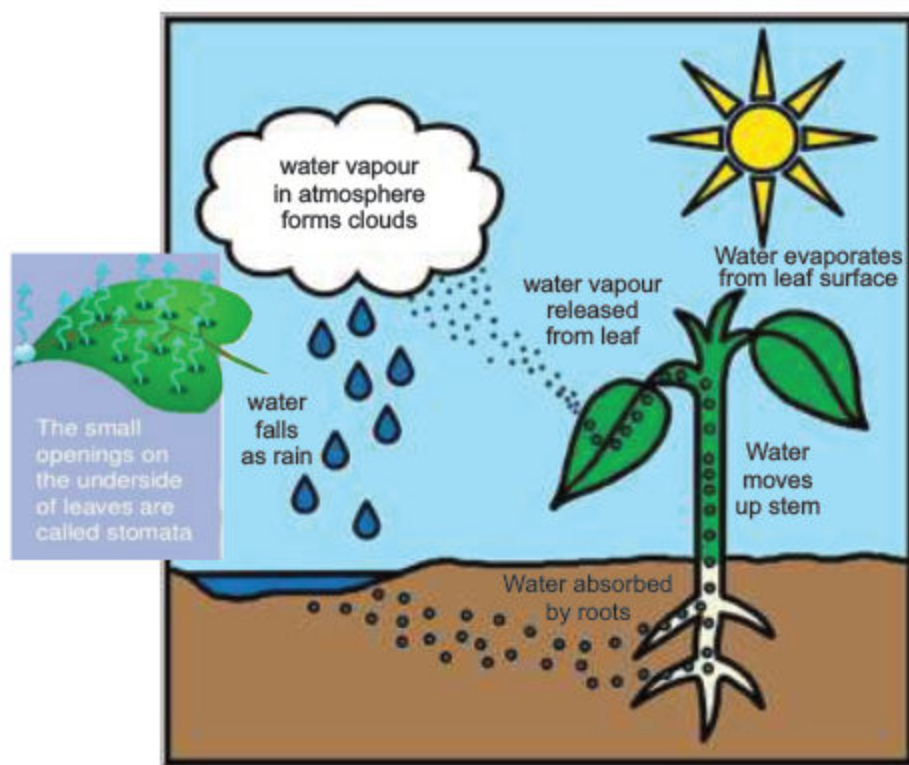


Fig: 9.4 Transpiration – showing loss of water vapours from stomata

Evidences of transpiration:

- Take a potted plant and wrap a polythelene bag around pot not around the plant to make sure that water is not coming from soil of pot and surface of pot.

- Place the pot on the glass plate and cover it with dry bell jar.
- Take another jar without plant for control setup.
- Put these two jars side by side in an area where light fall on it for two hours.

Observation:

You will observe water droplets at the bell jar which has plant while other jar remains dry.

9.2.1 Relation of transpiration with leaf surface:

Plants have three types of leaves on the basis of stomatal distribution.

- (i) Leaves that have stomata at lower epidermis called **bifacial leaves** e.g leaves of mango plant.
- (ii) Leaves that have stomata at both surfaces (upper and lower epidermis) called **monofacial leaves** e.g leaves of maze plant.
- (iii) Leaves that have stomata at upper epidermis only e.g leaves of water lily.

Activity: To find that transpiration mainly take place through stomata perform simple experiment.

Requirement:

Few leaves, petroleum jelly or wax, scale etc.

Procedure:

- Take three leaves of peapal or mango tree where stomata are mainly present at lower surface of leaves.
- The leaves should be of equal size.
- Treat leaves as follows:
Leaf 1- cover the upper epidermis surface with petroleum jelly or any wax.
Leaf 2- cover lower surface with same.
Leaf 3- cover both surfaces with same.
- Weigh each leaf before and after covering.
- Hang these leaves near window in sunlight.
- After few hours note the conditions and weigh again.
- Leaf which loss more weight transpire effeciently.
- It will be observed that the leaf 1 transpire more effeciently because it has stomata at lower epidermis.

The experiment shows that most of the water vapour is lost from the surface where stomata are present.

Surface area of leaf is also an important factor for the rate of transpiration, because the larger size leaves have high number of stomata which increases the rate of transpiration. The desert plants require to save their water so they have smaller size leaves or their leaves become spines to reduce number of stomata as well as rate of transpiration.

9.2.2 Stomata and its opening/closing mechanism:

Stomata are the pores usually found in the leaf epidermis surrounded by two kidney shaped guard cells, these cells contain chloroplast while other epidermal cells do not. The guard cells control the opening and closing of stomata. The inner wall of guard cells is thick and inelastic whereas the outer wall is thin, elastic and permeable. The changes in the turgidity of guard cells controls the opening and closing of stomata. Stomata open when the guard cell become turgid and close when the guard cells become flaccid. The turgidity of guard cell is regulated by concentration of solutes present in it which mainly depends upon the rate of photosynthesis. Opening and closing of stomata is one of the important factor to control rate of transpiration. The stomata remain open during the sunny day, as a result rate of transpiration increases. But at night they are closed, hence transpiration also stops.

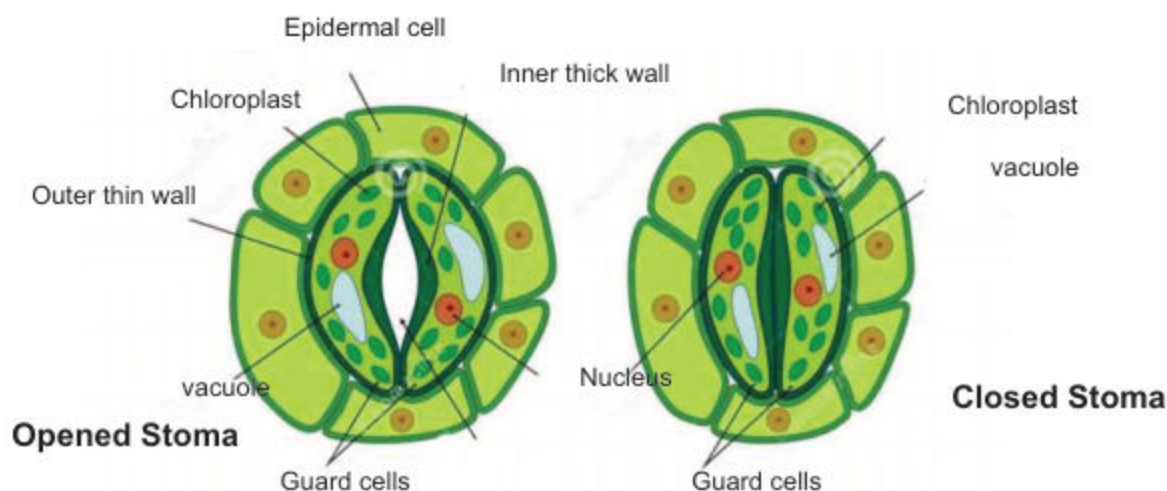


Fig: 9.5 Diagram showing opening and closing of stoma

9.2.3 Significance of transpiration:

Transpiration maintains low concentration of water and high concentration of solutes in cell i.e. high solute potential. The high solute potential of leaf cells attract more water and draw more water from xylem. This continuous withdrawal of water from xylem develops deficit of water in xylem which develops a pull or tension called transpiration pull. As a result of this transpiration pull and water attraction for other water molecule i.e cohesion of water, water is pulled upward in the xylem vessels through a continuous column called transpiration stream, which helps in ascent of sap.

- By active transpiration, transpiration pull is created which helps in the ascent of sap.
- Transpiration also increases the rate of absorption because the loss at one end increases demand on other end.
- Transpiration gets rid of the excess amount of water from plant.
- Transpiration helps in maintaining the temperature of plant for its metabolism and survival because evaporation causes cooling.
- Opening and closing of stomata is also regulated by transpiration, which indirectly influences upon the rate of photosynthesis and respiration.

Every year number of plants die in hot summer day due to excessive loss of water from aerial parts which results in wilting and dehydration and ultimately may lead to death of plant in extreme conditions.

Transpiration is considered as necessary element for plant due to its advantages which are discussed above but on the other hand, it is also considered as an evil for plant because million of plants die every year due to excessive transpiration.

9.2.4 Factors affecting the rate of transpiration:

The rate of transpiration is also affected by some of the following environmental factors.

- (i) **Temperature:** Rate of evaporation of water from cell surface increases with increase in temperature.